CHAPTER 15 ELECTROMAGNETIC SPECTRUM

3.3 Electromagnetic spectrum

1 Know the main regions of the electromagnetic spectrum in order of frequency and in order of wavelength

2 Know that the speed of all electromagnetic waves in:

(a) a vacuum is 3.0×10^8 m / s

(b) air is approximately the same as in a vacuum

3 Describe the role of the following components in the stated applications:

(a) radio waves – radio and television communications, astronomy

(b) microwaves – satellite television, mobile (cell) phone, Bluetooth, microwave ovens

(c) infrared – household electrical appliances, remote controllers, intruder alarms, thermal imaging, optical fibres

(d) visible light – photography, vision

(e) ultraviolet – security marking, detecting counterfeit bank notes, sterilising water

(f) X-rays – hospital use in medical imaging, security scanners, killing cancerous cells, engineering applications such as detecting cracks in metal

(g) gamma rays – medical treatment in detecting and killing cancerous cells, sterilising food and medical equipment, engineering applications such as detecting cracks in metal

4 Describe the damage caused by electromagnetic radiation, including:

(a) excessive exposure causing heating of soft tissues and burns

(b) ionising effects caused by ultraviolet (skin cancer and cataracts), X-rays and gamma rays (cell mutation and cancer)

1. O/N/23/12/Q30

Ultraviolet radiation is a component of the electromagnetic spectrum.

Which application uses ultraviolet radiation?

A Bluetooth technology

B prenatal scanning

C sterilising water

D thermal imaging

2. M/J/23/11/Q23

Which row names a region of the electromagnetic spectrum with a wavelength longer than that of visible light and a region with a frequency that is greater than that of visible light?

	wavelength longer than visible light	frequency greater than visible light
A	infrared	ultraviolet
B	microwave	radio
C	radio	microwave
D	ultraviolet	infrared

3. O/N/22/12/Q26

Many devices produce electromagnetic waves when operating.

Which device produces electromagnetic waves of the highest frequency?

A mobile phone

B sunbed

C television controller

D toaster

4. M/J/22/11/Q23

Which waves are **not** part of the electromagnetic spectrum?

A radio waves

B television signals

C ultrasound

D ultraviolet

5. O/N/21/11/Q24

Which component of the electromagnetic spectrum is used for the remote control of a television?

A gamma rays

B infrared rays

C radio waves

D ultraviolet rays

6. M/J/21/12/Q27

Which two waves are components of the electromagnetic spectrum?

A light and sound

B water waves and infrared

C ultrasound and ultraviolet

D X-rays and microwaves

7. O/N/20/12/Q25

Applications use different components of the electromagnetic spectrum.

Which shows correct applications for X-rays, ultraviolet light and microwaves?

	X-rays	ultraviolet light	microwaves
A	mobile phone	fluorescent tube	intruder alarm
B	killing cancerous cells	sterilising surgical instruments	satellite television
C	medical imaging	television controller	sunbed
D	sterilising surgical instruments	television controller	detecting cracks in metal

8. O/N/18/12/Q26

Which statement about radio waves is correct?

A Radio waves are sound waves.

B Radio waves are used to kill cancerous cells.

C Radio waves are used in television communications.

D Radio waves have frequencies higher than those of visible light.

9. M/J/18/12/Q30

A television controller emits an infra-red beam.

Which statement about infra-red radiation is correct?

A It causes ionisation.

B It consists of longitudinal waves.

C It has a higher frequency than ultra-violet light.

D It travels at the speed of light.

10. O/N/17/11/Q23

A type of electromagnetic radiation possesses the following properties.

- It is ionising.
- Its frequency is higher than the frequency of microwaves.
- It is not detected by the human eye.

What is this radiation?

A gamma rays

B infra-red

C light

D radio waves

11. O/N/17/12/Q24

The diagram shows the electromagnetic spectrum with three components named. The spectrum is in order from long wavelength to short wavelength.

Which component of the spectrum is used in a sunbed to produce a suntan?

long wavelength → short wavelength

radio waves	**A**	**B**	visible light	**C**	**D**	gamma rays

12. M/J/17/12/Q29

Which type of wave is used to send television signals to a satellite?

A infra-red waves

B light waves

C microwaves

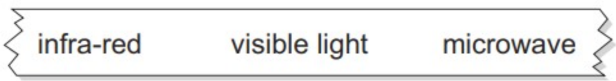
D sound waves

13. M/J/17/11/Q29

A wall poster showing the electromagnetic spectrum is displayed in a laboratory.

A section of the electromagnetic spectrum has been accidentally ripped from this wall poster.

Which piece is missing?

A 

B 

C 

D 

14. O/N/16/11/Q28

Which two types of electromagnetic radiation are both used to kill cancerous cells and are both used to detect cracks in metals?

A gamma-rays and microwaves

B gamma-rays and X-rays

C microwaves and ultra-violet radiation

D ultra-violet radiation and X-rays

15. M/J/16/11/Q24

Microwaves are used to transmit television signals to and from a satellite.

Which statement about microwaves is correct?

A They have a longer wavelength than radio waves.

B They penetrate the atmosphere without significant loss of energy.

C They travel much faster than radio waves in a vacuum.

D They warm the satellite and stop it freezing.

16. M/J/16/11/Q25

Where are gamma-rays used?

A in fluorescent tubes

B in killing cancerous cells

C in pre-natal scanning

D in sunbeds

17. M/J/15/12/Q28

Which component of the electromagnetic spectrum is used for television transmission from satellites?

A microwaves

B radio waves

C ultra-violet

D X-rays

18. O/N/14/11/Q24

Which application uses microwaves?

A detecting small cracks in metals

B gaining a sun-tan

C lighting a fluorescent tube

D satellite television

19. M/J/14/12/Q27

Which statement is correct for all electromagnetic waves?

A They are transverse.

B They cannot travel in a vacuum.

C They have the same frequency.

D They travel through lead.

20. M/J/14/11/Q26

Which statement is correct?

A Gamma rays have a longer wavelength than ultra-violet waves.

B Infra-red waves have a lower frequency than radio waves.

C Microwaves have a longer wavelength than visible light.

D X-rays have a higher speed in air than visible light.

21. O/N/13/11/Q22

The table lists the main components of the electromagnetic spectrum and their approximate frequency range.

	frequency / Hz
gamma rays	10^{22} to 10^{19}
X-rays	10^{21} to 10^{18}
ultra-violet	10^{18} to 10^{15}
visible light	10^{15} to 10^{14}
infra-red	10^{14} to 10^{12}
microwaves	10^{12} to 10^{9}
radiowaves	10^{9} to 10^{3}

Which range of frequencies can be used to detect cracks inside a block of metal?

A 10^{3} Hz to 10^{12} Hz

B 10^{12} Hz to 10^{15} Hz

C 10^{15} Hz to 10^{18} Hz

D 10^{18} Hz to 10^{22} Hz

22. M/J/13/11/Q22

Which wave property has the same value for all X-rays travelling in air?

A amplitude

B frequency

C speed

D wavelength

23. M/J/12/12/Q25

Which device uses ultra-violet radiation?

A electric grill

B intruder alarm

C television remote controller

D sunbed

24. M/J/12/11/Q25

Below are four statements about the uses of electromagnetic radiation.

Gamma rays are used in medical treatment.

Infra-red waves are used in sunbeds.

Microwaves are used in satellite television.

X-rays are used in intruder alarms.

How many of these statements are correct?

A 1 **B** 2 **C** 3 **D** 4

25. O/N/11/11/Q24

The diagram shows the main sections of the electromagnetic spectrum in order of increasing frequency. Some of the sections are labelled.

radio waves		R	light			gamma-rays

→ increasing frequency

The section R has a frequency just below that of light.

Which application uses the section R?

A killing cancerous cells

B satellite television

C sterilisation

D television remote controller

26. M/J/11/11/Q24

Which application may use the part of the electromagnetic spectrum called microwaves?

A cooking vegetables

B detecting small cracks in metals

C gaining a sun-tan

D lighting a fluorescent tube

27. O/N/10/11/Q24

Which pair of emissions travels with the same speed in air?

A alpha-particles and gamma-rays

B gamma-rays and infra-red waves

C infra-red waves and sound waves

D sound waves and alpha-particles

28. O/N/07/Q22

Which does **not** normally use infra-red radiation?

A electric grill

B intruder alarm

C television remote controller

D sunbed

29. M/J/06/Q25

Which wave is part of the electromagnetic spectrum?

	speed / m/s	type
A	330	longitudinal
B	330	transverse
C	3×10^8	longitudinal
D	3×10^8	transverse